LFG-Net: Low-Level Feature Guided Network for Precise Ship Instance Segmentation in SAR Images

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***Abstract*— Ship instance segmentation of high-resolution (HR) synthetic aperture radar (SAR) images is a valuable and challenging task due to the complex scattering and noise properties. In this article, we pioneered the construction of the low-level feature to discriminate the ships and complemented the super-resolution (SR) denoising techniques in the network modules, termed low-level feature guided network (LFG-Net), for precise ship instance segmentation in SAR images. LFG-Net consists of the low-level feature concerned pyramid (LFCP), the high-resolution feature interaction module (HR-FIM), and the compression recovery segmentation branch (CRSB). LFCP extends the vanilla feature pyramid network (FPN) with the P_1 layer and complements SR techniques to capture the regional and texture information at the image level for small object segmentation. HR-FIM interacts with the bounding box region of interest (RoI) feature and mask RoI feature at the instance level with HR techniques to enhance the mask RoI feature. CRSB aims at recovering the HR mask predictions to improve the ship segmentation performance. Comprehensive experiments on high-resolution SAR images dataset (HRSID), polygon segmentation SAR ship detection dataset (PSeg-SSDD), and AirSARShip indicate that LFG-Net* achieves 11.7%, 6.3%, and 12.7% AP increments compared with the Mask R-CNN baseline, respectively. Besides, it receives 9.5%, 4.9%, and 7.3% AP increments compared with the state-of-the-art method, which bridges the gap of instance segmentation precision in SAR images. In terms of the visualized instance segmentation results, LFG-Net* is capable of segmenting the complex scenes, e.g., the adjacent distributed ships and ships with strong reflection noise interference, in SAR images. The code is available at: https://github.com/Evarray/LFG-Net.**

***Index Terms*— Feature pyramid network (FPN), instance segmentation, region of interest (RoI), ship detection, synthetic aperture radar (SAR).**

I. Introduction

COMPARED with optical and hyperspectral images, synthetic aperture radar (SAR) images are more applicable to civilian fields (e.g., natural disaster forewarn and terrain evolution) and military fields (e.g., national defense and maritime transport safety) due to the capability of working in all-day and all-weather conditions. Nevertheless, influenced by the expensive measured SAR data, the data capacity of high-resolution (HR) SAR images is limited in the SAR community. In recent years, with the launch of SAR satellites, e.g., Sentinel-2 and Gaofen-3, a large amount of HR SAR data is freely available, which immensely promotes the visual application of SAR images. Specifically, in maritime domain awareness, researchers tend to apply airborne or spaceborne HR SAR images to automatic ship surveillance.

In the early days of SAR ship detection, the taxonomy of the detection methods can be categorized into single-feature-based methods, multifeature-based methods, and expert-system-oriented methods. Among them, constant false alarm rate (CFAR) detection methods [1], [2] have been widely explored and applied due to their signal processing properties (e.g., bandpass filter) and pattern recognition properties (e.g., one-class classifier). However, the defects, such as strong reflection area dependence and low detection speed, limit their application in automatic SAR ship detection. Fortunately, the advent of deep convolutional neural networks (DCNNs) brought new breakthroughs to automatic SAR ship detection. Namely, compared with traditional methods, DCNN-based methods generate a more considerable margin between robustness, detection speed, and end-to-end training ability. In the optical field, DCNN-based methods have been widely applied in the tasks of scene classification, semantic segmentation, and object detection [3]. He *et al.* [4] proposed the tracking via graph-based multitask reasoning (TGraM) to achieve efficient collaborative learning between detection and reidentification (ReID). Li *et al.* [5] combined the twin-auxiliary (TA) model and conflict reduction mechanism to achieve the class-incremental learning (CIL) network for semantic segmentation of small objects. Sun *et al.* [6] designed a unified part-based convolutional neural network (PBNet) for composite object detection. In the SAR field, DCNN-based methods have emerged in real-time, high-precision, and rotated bounding box-oriented ship detection. Kang *et al.* [7] regarded the region proposals generated by Faster R-CNN as the guard windows of traditional CFAR algorithm to detect the small ships in SAR images. Zhang and Zhang [8] designed the field-programmable gate arrays (FPGAs) embeddable SAR ship detector with depthwise separable convolution (DS-Conv). To detect ships under complex background, Wei *et al.* [9] proposed the HR-SDNet for precise and robust ship detection. Distinguished from prevalent ship detection

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methods based on horizontal bounding box, An *et al.* [10] used the oriented bounding box to generate the regions of interest (RoIs) for reducing the influence of background interference from areas outside the ship. Due to the low dependence on vision-specific inductive bias, vision transformer (ViT) has become the hot spot in computer vision (CV) [11]. Liu *et al.* [12] proposed a hierarchical transformer, termed Swin Transformer, to reduce the computation of self-attention with nonoverlapping local windows. Qu *et al.* [13] applied the transformer encoder in the CNN-based network to model the context information of targets. Xia *et al.* [14] took advantage of global contextual information perception of ViTs and local feature representation capabilities of DCNNs for contextual joint-representation learning in SAR ship detection.

However, ship detection based on bounding boxes merely discriminates the location, not the refined contour features, of the ships, which makes the discrimination ability of ship detectors stagnate at the instance level. Elevated from instance-level to pixel-level discrimination, instance segmentation combines the characteristics of pixel-level segmentation within the object categories in semantic segmentation and the instance-level discrimination and location for each object in object detection, which preserves the contour and orientation of objects. Moreover, with the launch of commercial SAR satellites, e.g., ICEYE X-2 [15] (December 2018), Capella-4 [16] (January 2021), and Hisea-1 [17] (December 2020), lower cost of acquiring HR SAR images will prevail. The orientation, bow, shape, and so on of HR ships captured by instance segmentation can provide prior information for military strikes and have broad application prospects. In [18], we attempt to enhance the perception capability of RoI extractors in multiclass object instance segmentation under the proprietary properties of aerial images. Unfortunately, due to the small object characteristics of ships and scattering and noise properties of SAR images, state-of-the-art methods in CV encounter bottlenecks in segmenting the ships as shown in Table XI, and precise ship instance segmentation methods remain unexplored in the SAR field.

Reviewing our previous work in [19], we found that deeper backbone networks have a limited impact on both ship detection and ship instance segmentation tasks, which further proves that such a large receptive field is not in accordance with the size of ships. Based on the above analysis, we attempt to design our ship segmentation network from the perspective of super-resolution (SR) and small object segmentation.

In this article, we introduce the notion of instance segmentation in the SAR field and propose the dedicated network, termed low-level feature guided network (LFG-Net), whose main components include low-level feature concerned pyramid (LFCP), high-resolution feature interaction module (HR-FIM), and compression recovery segmentation branch (CRSB). Existing object detection and instance segmentation methods tend to leverage direct output $\boldsymbol{C_2}, \boldsymbol{C_3}, \boldsymbol{C_4}$, and $\boldsymbol{C_5}$ of the backbone network to build feature pyramid network (FPN) [20], thus causing small objects to be pooled in the $\boldsymbol{F_2}$ layer of FPN. However, the $\boldsymbol{F_2}$ layer is low resolution and lacks the regional and texture information for segmenting the small objects. Therefore, LFCP extends a lower $\boldsymbol{F_1}$ layer of FPN to match the receptive field of small ships, recovers the regional and texture information with SR techniques, and truncates the high-level information flow, thereby facilitating both ship detection and ship instance segmentation tasks with the low-level feature. In addition to the image-level features generated in LFCP, we further enhanced the instance-level RoI features by HR-FIM to guide the segmentation task. HR-FIM interacts with the bounding box detection feature and the mask segmentation feature and adaptively captures the regional and texture information of ships with HR-FRM. Finally, CRSB is responsible for recovering the HR predictions with the zigzag compression techniques of SAR images. It is worth noting that the proposed LFG-Net not only bridges the gap between ship detection and ship instance segmentation in top-down methods but also breaks the bottleneck of segmentation precision in state-of-the-art methods. Succinctly put, LFG-Net pioneered the process of ship instance segmentation from image level to instance level to vector level and achieved excellent results in segmenting the small ships.

Our contributions can be summarized as follows.

1) LFG-Net pioneered the utilization of low-level feature to capture the regional and texture information with SR techniques at the image level and instance level to match the receptive field of small ships.
2) LFG-Net greatly bridges the gap between ship detection and ship instance segmentation in top-down methods and breaks the bottleneck of segmentation precision in state-of-the-art methods.
3) Comparative and ablation experiments on high-resolution SAR images dataset (HRSID), polygon segmentation SAR ship detection dataset (PSeg-SSDD), and AirSARShip demonstrate that LFG-Net* receives 9.5%, 4.9%, and 7.3% AP increments compared to the state-of-the-art method, respectively.
4) LFG-Net* can precisely segment the complex scenes, e.g., the adjacent distributed ships and ships with strong reflection noise interference, and can suppress the inadequacies in the state-of-the-art method.

II. Related Work

A. Instance Segmentation

Analogous to object detection methods, instance segmentation methods can be mainly categorized into top-down and bottom-up methods.

Top-down instance segmentation methods follow the criterion of detect-then-segment. He *et al.* [21] paralleled a mask branch with the detection branch in Faster R-CNN, termed Mask R-CNN, for instance segmentation. Cai and Vasconcelos [22] proposed the Cascade Mask R-CNN, which implements the multistage cascade architecture with increasing intersection over union (IoU) for high-quality hypotheses prediction. Based on Cascade Mask R-CNN, Hybrid Task Cascade (HTC) [23] introduced the joint multistage processing and complemented the spatial context enhancement for precise mask prediction. To calibrate the mismatch between mask quality and mask score, Huang *et al.* [24] regressed the mask IoU with IoU prediction head. Path aggregation

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network (PANet) [25] transferred the localization signals from the bottom layer to the top layer in multilevel propagation of top-down methods. Kirillov *et al.* [26] regarded the image segmentation as a point-based rendering problem and achieved excellent results.

Counterpart bottom-up instance segmentation methods ignore the procedure of generating the region proposals and RoIs in top-down methods and finish the segmentation task in one shot. Yolact [27] generates the prototype mask and correlated mask coefficients for mask prediction, thereby achieving temporal stability. The SOLO series proposed by Wang *et al.* [28], [29] introduce the instance categories according to the location and size of each instance.

B. Small Object Segmentation

In oceanographic applications, ships of small size in spaceborne SAR images are the targets of interest. Correspondingly, small and medium ships defined by MS COCO [30] evaluation metrics dominate the majority of existing SAR datasets, such as HRSID [19] (98% ships) and SSDD [31] (97.3% ships). Hence, we are prone to enhancing the ship instance segmentation capabilities of the network from the perspective of small object characteristics. As the large scale facilitates the detection of small objects, implementing SR techniques in the segmentation network can capture the texture information of the SAR ships while filtering the signal noise. In general image SR, researchers have developed a variety of SR blocks/modules for feature upsampling or feature denosing. Wang *et al.* [32] proposed the Content-Aware ReAssembly of FEatures (CARAFE) for HR feature upsampling. Zhang *et al.* [33] adopted the residual channel attention block (RSAB) to capture the informative components. Zhang *et al.* [34] designed the neural texture transfer architecture to formulate the reference-based super-resolution (RefSR). Deng *et al.* [35] super-resolved the pyramid features and extracted credible regional details with feature texture transfer module. Gong *et al.* [36] proposed the effective fusion factor for adapting the top-down information flow of FPN in detecting tiny objects. In addition to the general SR techniques mentioned above, some studies leverage the generative adversarial network (GAN) to generate the super-resolved features or images of small objects for the detection task. Li *et al.* [37] developed the perceptual GAN to generate the narrowing feature representation difference of small objects from large objects. Bai *et al.* [38] proposed the multitask GAN to recover detailed information of small objects and generate super-resolved images for accurate detection. Noh *et al.* [39] introduced the task-driven SR techniques in small object detection and utilized the HR target feature to supervise the small RoIs of objects.

Inspired by the small object detection methods and SR techniques, we designed the super-resolution feature reconstitution module (SR-FRM) and HR-FRM in LFCP and HR-FIM, respectively. SR-FRM captures the detailed and discriminative feature representation of small ships with the super-resolved low-level feature. Also, HR-FIM fuses the super-resolved RoI feature at the instance level.

C. Segmentation Branch in Top-Down Methods

As top-down instance segmentation methods simultaneously present the characteristics of instancewise detection and pixelwise segmentation. Researchers tend to improve instance segmentation performance by improving the output mask quality of the segmentation branch. He *et al.* [21] pioneered extending the Faster R-CNN with a fully convolution network (FCN) in semantic segmentation, for instance segmentation. Cheng *et al.* [40] simultaneously predicted the boundary feature and mask feature and fused them to enrich the mask feature. Vu *et al.* [41] proposed the residual mask branch under the premise of cascaded detection architecture to achieve the training inference sample consistency for cascaded methods. Ding *et al.* [42] adopted the mask feature generated by the mask proposal network (MPN) for mask prediction where the predicted mask in the previous stage is applied to progressive RoIAlign. Cao *et al.* [43] utilized two deconvolution layers for mask upsampling and two fully connected layers for mask scoring to construct the segmentation branch. Liu *et al.* [25] presented the fully connected fusion to perceive the spatial location of mask prediction. Chen *et al.* [23] proposed the interweaved mask branch for multistage processing.

III. Methodology

In the nutshell, LFG-Net follows the top-down instance segmentation formula, and the network architecture is shown in Fig. 1. First, the backbone network (ResNet-50 [44]) is responsible for extracting the deep feature representation of the input SAR image. Second, the hierarchical feature maps (including the enhanced $\boldsymbol{F_1}$ layer) provided by LFCP are sent to a single RoI extractor for generating the bounding box RoI feature and mask RoI feature. Third, the flattened bounding box RoI feature interacts with the mask RoI feature via HR-FIM. Finally, the enhanced mask RoI feature is transmitted to CRSB for mask prediction. Details are provided in Algorithm 1.

A. Low-Level Feature Concerned Pyramid

1) Vanilla FPN Architecture: Vanilla FPN extracts the feature level $\boldsymbol{C_2}, \boldsymbol{C_3}, \boldsymbol{C_4}$, and $\boldsymbol{C_5}$ (with the scale of $1/4^2, 1/8^2, 1/16^2$, and $1/32^2$ to the input image) from the backbone network to construct the basic hierarchical feature layers. Then, the adjacent feature layers are aggregated to build the output feature maps $\boldsymbol{P_2}$, $\boldsymbol{P_3}$, and $\boldsymbol{P_4}$

$$\boldsymbol{P_i} = f_{\text{out}_i}\left(f_{\text{laterals}_i}(\boldsymbol{C_i}) + f_{\text{upsample}_i}(\boldsymbol{C_{i+1}})\right) \tag{1}$$

where f_{laterals} and f_{out} are the 1×1 convolution kernel and 3×3 convolution kernel for channel reduction and feature processing, respectively, f_{upsample} is the nearest neighbor interpolation for top-down feature upsampling, and $\boldsymbol{P_5}$ layer is equal to $\boldsymbol{C_5}$ layer. Finally, an additional level $\boldsymbol{P_6}$ is downsampled from $\boldsymbol{P_5}$ via maxpool operation for detecting the large objects. In the subsequent procedure of RoI extraction, the corresponding ith FPN layer ($\boldsymbol{P_i}$) for RoIAlign is defined by the scale S of the region proposal (from RPN [45])

$$i = \lfloor \log_2\left(\sqrt{S}/56\right) \rfloor + 2. \tag{2}$$

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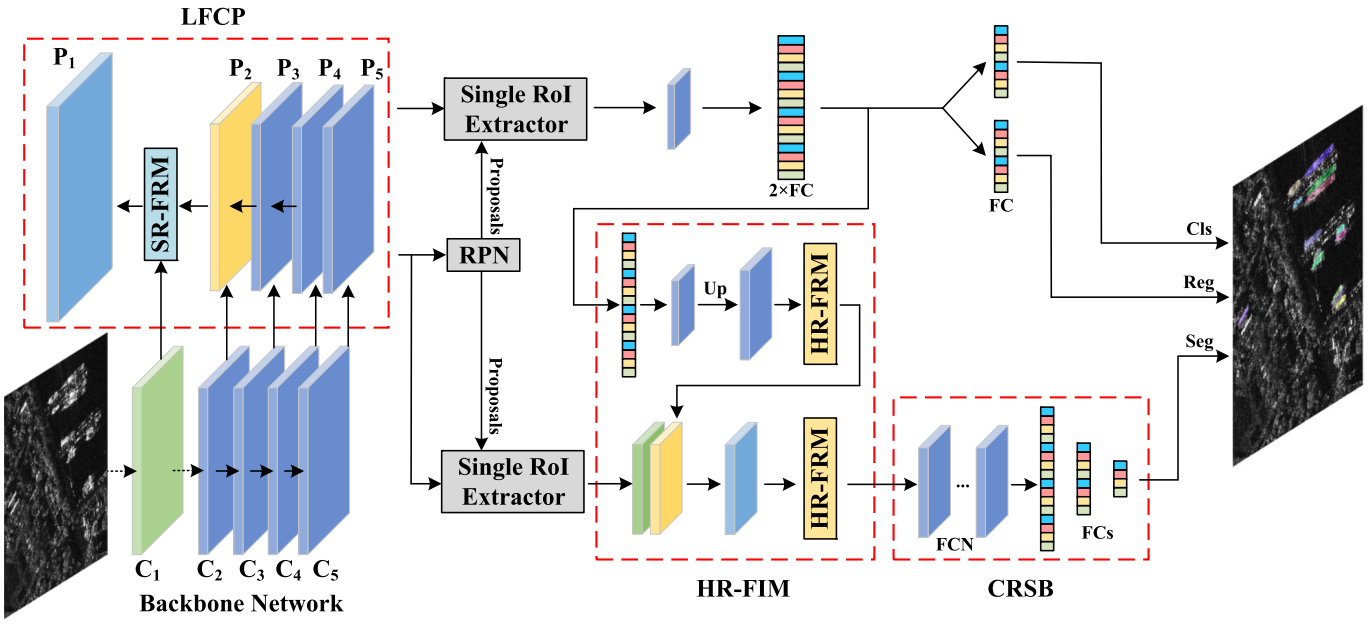

Fig. 1. Overview network architecture of LFG-Net, which follows the top-down instance segmentation formula. It replaces vanilla FPN with LFCP and supplements the HR-FIM and CRSB to boost the ship segmentation performance.

Algorithm 1 LFG-Net for SAR Ship Instance Segmentation

Given: Gray SAR images $\mathbf{X} = \sum_{i=1}^{n} \mathbf{x_i}$, $\mathbf{x_i} \in \mathbb{R}^{H \times W \times 3}$;
Output: $\mathbf{Y} = \sum_{i=1}^{n} \mathbf{y_i}$, $\mathbf{y_i} \in \mathbb{R}^{H \times W \times 3}$, with ship segmentation mask $\mathbf{M}$, score $\mathcal{S}$, and location coordinates $\mathcal{C}$.

```
Dataset ∑X loading, hyperparamaters setting;
Extract hierarchical feature map with the backbone network and LFCP, build texture enhanced P1 with SR-FRM, obtain P1, P2, P3, P4, P5;
Stage1 :
    RPN proposal generation, obtain RP;
    Generate bounding box RoI (BR) with Pi and RP;
    Retain the flattened BR for HR-FIM;
    SAR ship classification and bounding box regression.
Stage2 :
    Generate original mask RoI (MR) with Pi and RP;
    Implement HR interation for flattened BR and MR;
    Model the segmentation results with FCN and FCs;
    Zig-zag uncompression and 2D-DCT transform;
    HR pixel-wise segmentation, S = Branch(P).
Do result mapping and integration, obtain Y.
```

However, this allocation strategy prompts the $\boldsymbol{P_2}$ layer of FPN to pool the objects with the scale under 112^2 pixels, which covers the small (scale $< 32^2$), medium ($32^2 <$ scale $< 96^2$), and large objects (scale $> 96^2$) in COCO evaluation metrics. As the overwhelming majority of ships in SAR images have scales below 96^2 pixels, vanilla FPN has limited improvement on ship detection, especially ship instance segmentation. Motivated by the analysis above, we proposed the LFCP to boost the performance of SAR ship segmentation by the super-resolved $\boldsymbol{P_1}$ layer.

2) LFCP Network Architecture: Low-level features contain more regional and texture information, which is beneficial to pixelwise segmentation tasks. Instead of applying the $\boldsymbol{P_2}$ layer to predict the small ships as vanilla FPN, LFCP extracts features from the network stem and fuses the $2\times$ upsampled intermediate $\boldsymbol{P_2}$ layer ($\boldsymbol{P_2^m}$) to build the output $\boldsymbol{P_1}$ layer. Assuming that the input image of the backbone network is $\boldsymbol{I}$, low-level feature from the backbone is generated by

$$\boldsymbol{F}_{\text{low}} = \text{Conv}_{7\times7}(\boldsymbol{I}; \theta)_{s=2;\, p=3} \tag{3}$$

where $\text{Conv}_{7\times7}(*; \theta)$ denotes the 7×7 convolution kernel with the weight θ. Distinguished from the top-down feature fusion criterion in vanilla FPN, we truncate the information of intermediate $\boldsymbol{P_5}$ layer ($\boldsymbol{P_5^m}$) and retain the information flow of intermediate $\boldsymbol{P_4}$ and $\boldsymbol{P_3}$ layer ($\boldsymbol{P_3^m}$, $\boldsymbol{P_4^m}$) to generate the intermediate $\boldsymbol{P_2}$ layer ($\boldsymbol{P_2^m}$). The process is formulated by

$$\boldsymbol{P_3^m} = f_{\text{laterals}}(\boldsymbol{C_3}) + f_{\text{nearest}}(f_{\text{laterals}}(\boldsymbol{C_4})) \uparrow_{2\times} \tag{4}$$

$$\boldsymbol{P_2^m} = f_{\text{laterals}}(\boldsymbol{C_2}) + f_{\text{nearest}}\left(\boldsymbol{P_3^m}\right) \uparrow_{2\times} \tag{5}$$

where f_{nearest} denotes the nearest interpolation. For the sake of aggregating the contextual information to prompt the pixelwise prediction, we generate the $\boldsymbol{P_2^m}$ indexed content-aware reassembly kernel ($\boldsymbol{K}$) [32] via

$$\boldsymbol{K} = \text{Softmax}\left(f_{ps}\left(\text{Conv}_{3\times3}\left(\boldsymbol{P_2^m}; \theta_1\right)\right) \uparrow_{2\times}\right) \tag{6}$$

where f_{ps} is the pixel shuffle operation and Softmax is the softmax function. Assuming that each location $w(x, y)$ in $\boldsymbol{P_2^m}$ is embraced by the pixel-centered region $\boldsymbol{Q}(P_{2w}^m, k)$, $\boldsymbol{Q}$ is synergistically elementwise multiplied with the counterpart location $w'(x', y')$ in $\boldsymbol{K}$ as the channel of $\boldsymbol{K}$ is k^2. The upsample process is formulated as follows:

$$\boldsymbol{P}_{2w'}^{up} = \sum_{n=-2}^{2} \sum_{n=-2}^{2} \boldsymbol{P_2^m}_{(i+n, j+m)} \cdot \boldsymbol{K}_{w'(n,m)} \tag{7}$$

where $\cdot$ denotes the elementwise multiplication.

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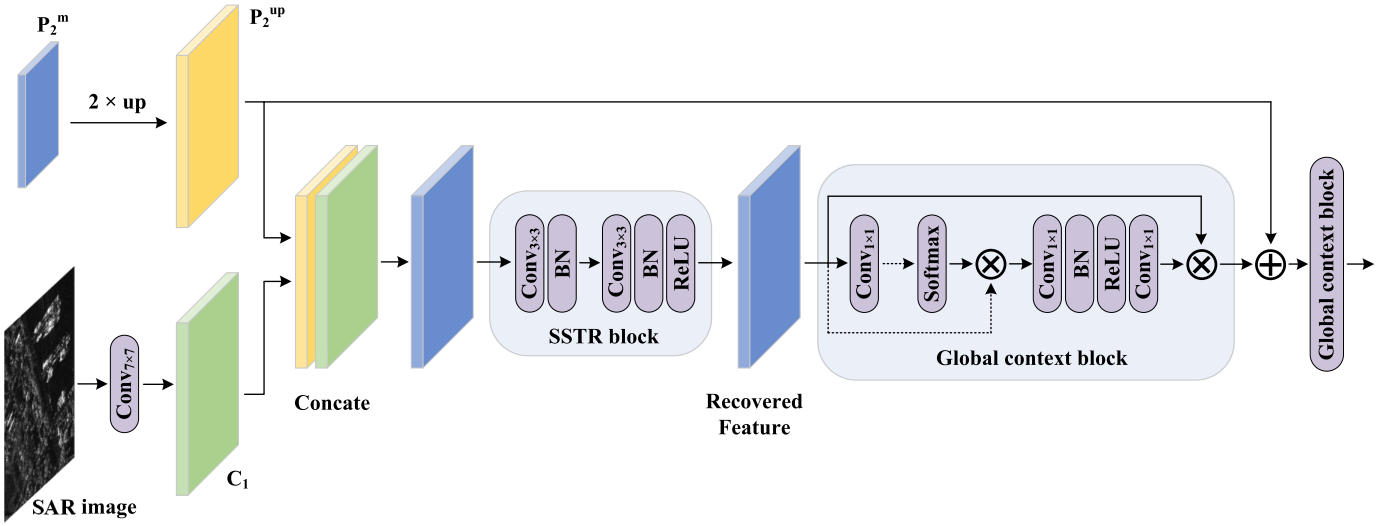

Fig. 2. Detailed architecture of the SR-FRM, where the dotted arrow denotes the multidimensional scaling.

As the low-level features $\boldsymbol{F}_{\text{low}}$ and $\boldsymbol{P}_2^{up}$ retain the 1/2 width and height to the input image $\boldsymbol{I}$, we proposed the SR-FRM to build the low-level feature. The detailed architecture of SR-FRM is shown in Fig. 2. First, we fuse $\boldsymbol{F}_{\text{low}}$ and $\boldsymbol{P}_2^{up}$

$$\boldsymbol{P}_1^m = \text{Conv}_{1\times1}\left(\boldsymbol{P}_2^{up} || \boldsymbol{F}_{\text{low}}; \theta_2\right) \downarrow_{2\times}^{C} \tag{8}$$

where $||$ denotes the channelwise feature concatenation operation and $\downarrow_{2\times}^{C}$ is the $2\times$ channel reduction. Inspired by the network design strategies in SR frameworks, we designed the SAR ship texture recovery (SSTR) block with residual learning, which is formulated as follows:

$$\boldsymbol{P}_1^{r'm} = \text{BN}\left(\text{Conv}_{3\times3}\left(\boldsymbol{P}_1^m; \gamma\right)\right) \tag{9}$$

$$\boldsymbol{P}_1^{rm} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(\boldsymbol{P}_1^{r'm}; \gamma_1\right)\right)\right) \tag{10}$$

where γ and γ_1 are the weights of 3×3 convolution kernel and $\boldsymbol{P}_1^{rm}$ is the recovered output feature. To capture the long-range spatial contextual information while modeling the channel interdependence of the feature, we adopt the global context block to enhance $\boldsymbol{P}_1^{rm}$ and build the intermediate $\boldsymbol{P}_1$ layer ($\boldsymbol{P}_1^m$) via

$$\boldsymbol{P}_1^m = f_{gc}\left(f_{gc}\left(\boldsymbol{P}_1^{rm}\right) + \boldsymbol{P}_2^{up}\right) \tag{11}$$

where f_{gc} denotes the global context block that maps the input feature $\boldsymbol{x}$ to the output $\boldsymbol{x_{gc}}$. First, we model the long-range spatial contextual information of the input feature ($\boldsymbol{x} \in \mathbb{R}^{C\times H\times W}$)

$$\boldsymbol{x_s} = f_{\text{reshape}}(x), \quad \boldsymbol{x_s} \in \mathbb{R}^{C\times HW} \tag{12}$$

$$\boldsymbol{x_w} = f_{\text{reshape}}\left(\text{Conv}_{1\times1}(\boldsymbol{x}; \gamma_2)|_1^C\right), \quad \boldsymbol{x_w} \in \mathbb{R}^{HW\times1\times1} \tag{13}$$

$$\boldsymbol{x_c} = \boldsymbol{x_s} * \text{Softmax}(\boldsymbol{x_w}), \quad \boldsymbol{x_c} \in \mathbb{R}^{C\times1\times1} \tag{14}$$

where γ_2 and $*$ denote the weight for 1×1 convolution kernel and matrix multiplication, respectively. Second, the channel interdependence of $\boldsymbol{x_c}$ is enhanced by the channel attention (squeeze excitation operation with the ratio r)

$$\boldsymbol{x}_{\boldsymbol{gc}}^{\boldsymbol{s}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}(\boldsymbol{x_c}; \gamma_3) \downarrow_{r\times}^{C}\right)\right) \tag{15}$$

$$\boldsymbol{x_{gc}} = x \cdot \text{Conv}_{1\times1}\left(\boldsymbol{x}_{\boldsymbol{gc}}^{\boldsymbol{s}}; \gamma_4\right) \uparrow_{r\times}^{C} \tag{16}$$

where γ_3 and γ_4 are the weights of 1×1 convolution kernel.

Distinguished from the output $\boldsymbol{P_6}$, $\boldsymbol{P_5}$, $\boldsymbol{P_4}$, $\boldsymbol{P_3}$, and $\boldsymbol{P_2}$ layers in vanilla FPN, LFCP leverages the lower feature layers to segment the small-sized ships. Given the intermediate feature layer $\boldsymbol{P}_5^m$, $\boldsymbol{P}_4^m$, $\boldsymbol{P}_3^m$, $\boldsymbol{P}_2^m$, and $\boldsymbol{P}_1^m$, LFCP builds the output $\boldsymbol{P_5}$, $\boldsymbol{P_4}$, $\boldsymbol{P_3}$, $\boldsymbol{P_2}$, and $\boldsymbol{P_1}$ layers via

$$\boldsymbol{P_i} = \text{Conv}_{3\times3}\left(\boldsymbol{P}_i^m; \delta_i\right), \quad \text{s.t. } i \in \{1,2,3,4,5\} \tag{17}$$

where δ_i is the weight for 3×3 convolution kernel.

3) RPN Anchor Generation: In vanilla FPN, the $\boldsymbol{P_6}$ layer is constructed by implementing the maxpool operation on the $\boldsymbol{P_5}$ layer. Hence, RPN generates the anchors with the basic scale of 8, the ratio of $\{0.5, 1, 2\}$, and the stride of $\{4, 8, 16, 32, 64\}$ on $\{\boldsymbol{P_2}, \boldsymbol{P_3}, \boldsymbol{P_4}, \boldsymbol{P_5}, \boldsymbol{P_6}\}$ to predict the region proposals. To focus on the small object property of SAR ships, LFCP interrupts the construction of $\boldsymbol{P_6}$ layer and enables RPN to generate the anchors with the basic scale of 8, the ratio of $\{0.5, 1, 2\}$, and the stride of $\{2, 4, 8, 16, 32\}$ on $\{\boldsymbol{P_1}, \boldsymbol{P_2}, \boldsymbol{P_3}, \boldsymbol{P_4}, \boldsymbol{P_5}\}$ instead.

B. HR-FIM

In vanilla Mask R-CNN, the pooled RoI features of the detection branch and mask branch are independently responsible for parallel bounding box regression and mask prediction tasks. However, the interaction of multitask branches is especially beneficial to pixelwise tasks. Therefore, we constructed the information flow from the ship detection branch to mask segmentation branch via the HR-FIM, as shown in Fig. 3.

Assuming that the pooled RoI feature for the detection branch is $\boldsymbol{RoI_D} \in \mathbb{R}^{N\times C\times7\times7}$, it is flattened and transferred to two shared fully connected layers in the detection branch

$$\boldsymbol{R_D} = \text{ReLU}(\text{FC}_{1024}(\text{ReLU}(\text{FC}(f_{\text{flatten}}(\boldsymbol{RoI_D}))))) \tag{18}$$

where ReLU and FC denote the activation function and fully connected layer, respectively, and $\boldsymbol{R_D}$ is the vanilla detection RoI feature for feature interaction. Since $\boldsymbol{R_D} \in \mathbb{R}^{N\times1024}$ is the intermediate feature of multitasking branches, the negative RoIs of ships are selectively filtered to match the mask branch

$$\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{v}} = f_{\text{slice}}(\boldsymbol{R_D}). \tag{19}$$

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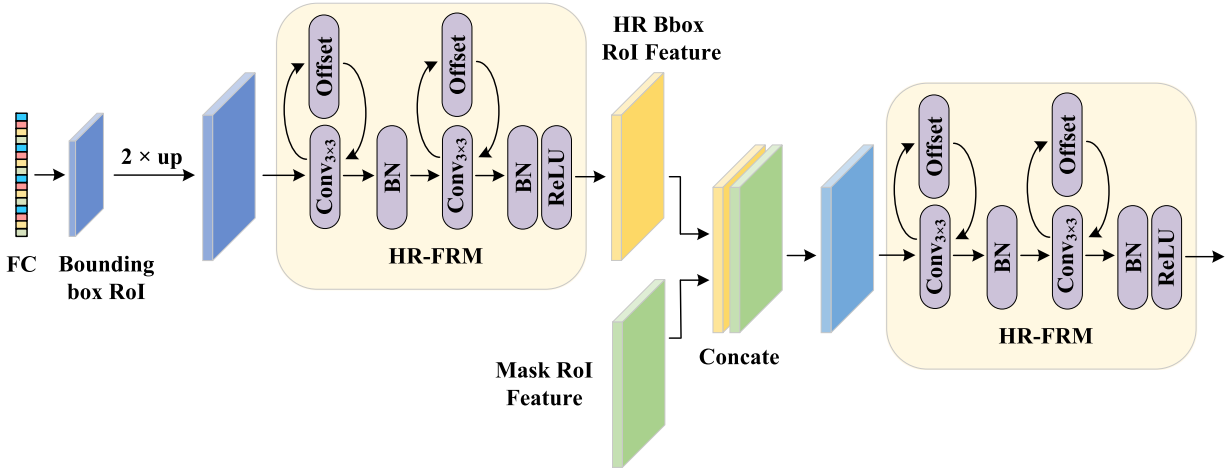

Fig. 3. Detailed architecture of the HR-FIM.

As $\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{v}}$ is the compressed vector, we recover it to generate the multidimensional RoI feature

$$\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm}} = f_{\text{reshape}}\left(\text{ReLU}\left(\text{FC}\left(\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{v}}\right)\right)\right) \tag{20}$$

where $\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm}} \in \mathbb{R}^{P\times C\times 7\times 7}$. Then, to keep the same size as the counterpart mask RoI feature, we further refactor it via bilinear interpolation function

$$\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm'}} = f_{\text{bilinear}}\left(\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm}}\right) \uparrow_{2\times} . \tag{21}$$

Since $\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm'}}$ is the objectwise feature in essence, we capture its pixelwise long-range dependencies via the HR-FRM presented in the following equations:

$$\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{m'}} = \text{BN}\left(\text{Conv}_{3\times 3}^{\text{OS}}\left(\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{bm'}}; w; os\right)\right) \tag{22}$$

$$\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{m}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times 3}^{\text{OS}}\left(\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{m'}}; w_1; os_1\right)\right)\right) \tag{23}$$

where $\text{Conv}_{3\times 3}^{\text{OS}}(*; w; os)$ denotes the 3×3 convolution kernel attached with the offset to model the objectwise geometric transformations within pertinent RoI; w and os are the weights for convolution kernel and offset, respectively; and BN represents the batch normalization.

Next, we fuse the reconstructed SAR ship detection RoI feature with the pooled mask RoI feature via

$$\boldsymbol{R}_{\boldsymbol{S}} = \text{Conv}_{1\times 1}\left(\boldsymbol{R}_{\boldsymbol{D}}^{\boldsymbol{m}} || \boldsymbol{R}_{\boldsymbol{M}}; w_2\right) \downarrow_{2\times}^{C} \tag{24}$$

where $||$ is the channelwise feature concatenation operation and $\text{Conv}_{1\times 1}(*; w_2)$ denotes the 1×1 convolution with the weight w_2 for feature dimensionality reduction.

Finally, we adopt the HR-FRM once again to acquire the HR interacted mask RoI feature

$$\boldsymbol{R}_{\boldsymbol{S}}^{\boldsymbol{H}} = f_{\text{HR-FRM}}(\boldsymbol{R}_{\boldsymbol{S}}) \tag{25}$$

where the function $f_{\text{HR-FRM}}$ is equivalent to the HR-FRM operation in (22) and (23).

C. Compress Recovery Segmentation Branch

1) Compress Recovery: In vanilla mask branch of Mask R-CNN, the input RoI feature $\in \mathbb{R}^{P\times 256\times 14\times 14}$ is $\uparrow_{2\times}$ unsampled to acquire the prediction and calculate the segmentation loss. However, calculating the pixelwise discrepancy within such a small region is prone to producing deviations. In this section, we introduce the zigzag compression [46] and discrete cosine transform (DCT) [47] to calculate the segmentation loss in the compressed vector dimension and recover it to the appropriate image size for mask prediction of the ships. Assuming that the 2-D size of the input image is $\boldsymbol{Q} \in \mathbb{R}^{S\times S}$, the 2-D DCT is formulated as follows:

$$\boldsymbol{Q}(x,y)_{\text{DCT}} = \frac{2}{S}\cdot C(x)C(y)\sum_{i=0}^{S-1}\sum_{j=0}^{S-1}\boldsymbol{Q}(i,j)\cdot \boldsymbol{D}_{x,y}^{i,j}$$
$$\text{s.t. } i \in \{0,1,\ldots,S-1\},\ \ j \in \{0,1,\ldots,S-1\} \tag{26}$$

$$D_{x,y}^{i,j} = \cos\left(\frac{\pi x(2i+1)}{2S}\right)\cos\left(\frac{\pi y(2j+1)}{2S}\right) \tag{27}$$

where $\boldsymbol{Q}(x,y)$ is the pixel located at (x,y) of the 2-D DCT output; $C(*) = 1/(2)^{1/2}$ while $* = 0$, and $C(*) = 1$ otherwise. Correspondingly, the 2-D inverse DCT (2-D IDCT) is formulated as follows:

$$\boldsymbol{Q}(i,j)_{\text{IDCT}} = \frac{2}{S}\cdot C(x)C(y)\sum_{x=0}^{S-1}\sum_{y=0}^{S-1}\boldsymbol{Q}(x,y)_{\text{DCT}}\cdot \boldsymbol{D}_{x,y}^{i,j}$$
$$\text{s.t. } x \in \{0,1,\ldots,S-1\},\ \ y \in \{0,1,\ldots,S-1\}. \tag{28}$$

2) Network Architecture: In the sequel of the output mask RoI feature $\boldsymbol{R}_{\boldsymbol{S}}^{\boldsymbol{H}}$ in Section III-B, it serves as the input of the CRSB. Consistent with the mask branch in Mask R-CNN, we follow [48] and adopt FCN to extract the deep mask representation:

$$\boldsymbol{M}_{\boldsymbol{S}} = \left(\text{ReLU}\left(\text{Conv}_{3\times 3}\left(\boldsymbol{R}_{\boldsymbol{S}}^{\boldsymbol{H}}; \xi\right)\right)\right)^{4\times} \tag{29}$$

where $4\times$ denotes the four times repeated operation and ξ is the weight for the 3×3 convolution. As for SAR ship segmentation, the channels of $\boldsymbol{M}_{\boldsymbol{S}}$ remain the same instead of being reduced to the number of classes. Instead, we compress $\boldsymbol{M}_{\boldsymbol{S}}$ into the compact vector via

$$\boldsymbol{V}_{\boldsymbol{pd}} = \text{FC}(\text{FC}_{1024}(\text{FC}(f_{\text{flatten}}(\boldsymbol{M}_{\boldsymbol{S}})))) \tag{30}$$

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where $\boldsymbol{V_{pd}} \in \mathbb{R}^{P\times D_{zigzag}}$ is the output for mask prediction. Here, the operation is equivalent to segmenting the SAR ships in the frequency domain. Correspondingly, the ground-truth mask is encoded from the signal domain to the frequency domain via 2-D DCT transform

$$\boldsymbol{M}_{\boldsymbol{gt}}^{\text{fre}} = f_{\text{DCT}}\left(\boldsymbol{M}_{\boldsymbol{gt}}\right) \tag{31}$$

where $\boldsymbol{M}_{\boldsymbol{gt}}^{\boldsymbol{fre}} \in \mathbb{R}^{P\times 1\times S_{gt}\times S_{gt}}$ is the frequency ground truth and f_{DCT} is the 2-D DCT operation in (26) and (27). Then, $\boldsymbol{M}_{\boldsymbol{gt}}^{\text{fre}}$ is compressed to a vector via zigzag algorithm

$$\boldsymbol{V}_{\boldsymbol{gt}} = f_{\text{zigzag}}\left(\boldsymbol{M}_{\boldsymbol{gt}}^{\text{fre}}\right). \tag{32}$$

As a result, we transform mask prediction task into a regression task. The prediction loss is calculated via mean absolute error (MAE)

$$\text{loss}_{\text{mask}} = \sum_{i=1}^{P} \frac{|\boldsymbol{V}_{\boldsymbol{pd}}^{\boldsymbol{i}} - \boldsymbol{V}_{\boldsymbol{gt}}^{\boldsymbol{i}}|}{P} \tag{33}$$

where $\boldsymbol{V}_{\boldsymbol{pd}}^{\boldsymbol{i}}$ and $\boldsymbol{V}_{\boldsymbol{gt}}^{\boldsymbol{i}}$ are the ith SAR ship in a batch. In the inference phase, the prediction V_{pd} is recovered by the 2-D IDCT transform in (28) and zigzag uncompression to get the SAR ship segmentation results.

IV. Experiments

A. Datasets

1) HRSID: HRSID originates from our previous work and it is widely implemented in developing ship detectors in the SAR field [49]–[53]. It contains 5604 HR SAR images with the resolution ranging from 0.5 to 3 m, and the images retain the size of 800×800 pixels. There are 16 951 ships in total with various polarization and imaging modes. HRSID is divided into the training set with 65% images and the test set with 35% images in our ship instance segmentation experiments, which is in accordance with the official scheme.

2) PSeg-SSDD: It is the extended version [31] of SSDD [54], which supports ship instance segmentation and ship detection tasks. It contains 1160 SAR images with a resolution ranging from 1 to 15 m, and there are 2456 ships in total. Distinguished from [31], PSeg-SSDD is divided into the training set with 65% images and the test set with 35% images in our experiments to reduce bias.

3) AIR-SARShip: Based on the prior bounding box annotations provided by Xian *et al.* [55], we relabel the ships in AIR-SARShip-v1.0 and AIR-SARShip-v2.0 dataset with polygons and extend it to MS COCO format to support both ship detection and ship instance segmentation tasks. The original AIR-SARShip dataset comprises 331 HR images with a resolution ranging from 1 to 3 m. Here, we crop the images into 600 × 600 pixels by a sliding window with a stride of 200 pixels and obtain 857 images (65% images) for training and 456 images (35% images) for testing. Consequently, the AIR-SARShip dataset contains 3463 ships in the training set and 1849 ships in the test set with various polarization and imaging modes.

B. Evaluation Metrics and Implementation Details

Akin to the object detection, ship instance segmentation selects the mask IoU as the basic criterion of evaluation

$$\text{IoU}_{\text{mask}} = \frac{\text{Pred}_m \cap \text{Gt}_m}{\text{Pred}_m \cup \text{Gt}_m} \tag{34}$$

where Pred_m is the mask prediction and Gt_m is the counterpart ground truth. Given a certain IoU threshold, the predictions can be categorized into true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Thus, we can define precision (P) and recall (R) via

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{35}$$

$$R = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{36}$$

Meanwhile, the threshold AP_{thr} is calculated by

$$AP_{\text{thr}} = \int_0^1 P dr. \tag{37}$$

In the sequel, AP is the mean value of 10 AP_{thr} value from the threshold of 0.5 to 0.95 with the stride of 0.05. In terms of the AP evaluation of multiscale objects, there are AP_S, AP_M, and AP_L in MS COCO evaluation metrics.

Besides, we supplement the parameters (denoted as Params.) and frames per second (FPS) to test the network complexity and test speed, respectively. Parameters are the criterion for measuring the number of learnable network parameters while training the CNN-based model. Assuming that the channel number of input feature is C_{in} and the size and output channels of convolution kernel are $M\times M$ and C_{out}, respectively, then parameters can be defined as

$$\text{Parameters} = \left(C_{\text{in}}M^2 + 1\right)C_{\text{out}}. \tag{38}$$

Apart from the evaluation metrics, we select the Pytorch framework for prototyping the models. A personal computer with Quadro RTX 6000 GPU (24-GB memory) is used for training and testing. In the training phase, the images in HRSID, PSeg-SSDD, and Air-SARShip are resized to 800 × 800 pixels, 512 × 512 pixels, and 600 × 600 pixels, respectively. The stochastic gradient descent (SGD) with the momentum of 0.9 and the weight decay of 0.0001 is used for optimization. The models are trained for 12 epochs in HRSID and PSeg-SSDD in total, where the initial learning rate is set at 0.0025 and decreased ten times at the 8th epoch and 11th epoch. In particular, we train the model for 36 epochs in AirSARShip with the initial learning rate of 0.0025 and ten times attenuation in the 24th epoch and 33rd epoch. In the test phase, we select the soft nonmaximum suppression (Soft NMS) [56] with the threshold of 0.5 to suppress the low score predictions in ship segmentation results.

C. Hypothesis Validation and Effects Analysis

As state-of-the-art methods encounter bottlenecks in segmenting ships in SAR images, we validate the previous hypothesis on SAR images in several aspects. Since ships maintain the characteristics of small objects, simply increasing

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TABLE I
EFFECTS OF NETWORK DEPTH ON SAR SHIP INSTANCE SEGMENTATION. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Backbone Network	Depth	AP	AP_{50}	AP_{75}
ResNet	50	52.2	80.5	63.1
	101	51.7	79.8	62.6
	152	52.6	80.2	63.9
ResNeSt	50	51.5	79.3	62.6
	101	51.5	78.5	62.9
	200	52.2	79.9	63.8
ResNeXt-32x4d	50	52.2	79.8	63.4
	101	52.7	88.2	64.2

TABLE II
EFFECTS OF IMAGE SIZE ON SAR SHIP INSTANCE SEGMENTATION. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Image Size	AP	AP_{50}	AP_{75}
600×600	45.1	72.0	53.9
800×800	52.2	80.5	63.1
1000×1000	54.8	85.6	65.8
1200×1200	56.7	87.4	68.2
1400×1400	**57.0**	**88.0**	**68.7**

the depth of the backbone network to enlarge the receptive field of multilevel feature maps limits the performance gain of ship instance segmentation. Hence, we select mainstream ResNet, ResNeSt [57], and ResNeXt [58] with Mask R-CNN architecture for experiments. As shown in Table I, AP value of ResNet, ResNeSt, and ResNeXt continuously fluctuates within 1% with the increasing network depth in SAR ship instance segmentation, which validates the necessity of low-level feature and diminishes the importance of high-level feature in ship instance segmentation. Hence, we simply select ResNet-50 as the benchmark backbone network for subsequent experiments. Higher resolution provides the details of small objects, which ensures precise segmentation of the SAR ships. As shown in Table II, we resized the SAR images from original 800×800 pixels to 600×600, 1000×1000, 1200×1200, and 1400×1400 pixels for experiments. Quantitatively, the segmentation AP converges at 1400×1400 pixels and reaches 57.0%. However, extra GPU memory usage and computation complexity for such a large input size are unaffordable in practical applications. Thus, we are prone to maintaining 800×800 pixels input and use SR techniques in constructing the essential modules of our LFG-Net.

We conduct experiments to test the modulewise utility of LFG-Net and the effectiveness of the ensemble model. As shown in Table III, LFCP, HR-FIM, and CRSB receive 6.0%, 6.4%, and 5.8% AP increments compared with the baseline (vanilla Mask R-CNN), respectively, where the results of HR-FIM and CRSB are based on LFCP-Basic. Without bells and whistles, our LFG-Net achieves the AP of 59.7%, which receives 7.5% AP increments compared with the baseline. Meanwhile, we visualize the qualitative SAR ship instance segmentation results for the cases of offshore ships, port ships, adjacent distributed ships, and so on in Fig. 4.

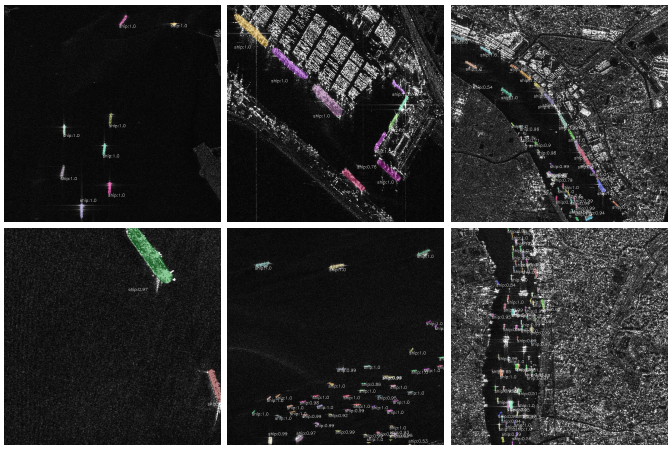

Fig. 4. Visualized SAR ship instance segmentation results.

D. Ablation Experiments

1) Experiments on LFCP: Inspired by the backbone network analysis in Section IV-C, we simply build the $\boldsymbol{F_1}$ level of FPN by adding $\boldsymbol{F_{low}}$ [in (3)] and the $2\times$ upsampled $\boldsymbol{P_2^m}$ layer [not in (5)]. Then, we envision whether higher layers in the top-down information flow of the FPN are beneficial for segmenting the SAR ships and truncate the information flow at a certain layer in our experiments. As shown in Table IV, with the regional and texture information provided by $\boldsymbol{F_1}$ level, all cases of information flow layers, including $\boldsymbol{F_5} \sim \boldsymbol{F_1}$, $\boldsymbol{F_4} \sim \boldsymbol{F_1}$, $\boldsymbol{F_3} \sim \boldsymbol{F_1}$, and $\boldsymbol{F_2} \sim \boldsymbol{F_1}$, receive higher AP values than vanilla FPN baseline (with the information flow layer of $\boldsymbol{F_5} \sim \boldsymbol{F_2}$). Even for the two-layer fusion case in $\boldsymbol{F_2} \sim \boldsymbol{F_1}$, it still yields 1.3% AP value than four-layer fusion case in $\boldsymbol{F_5} \sim \boldsymbol{F_2}$, which confirms the effectiveness of adding $\boldsymbol{F_1}$ layer in building our LFCP. Meanwhile, the four-layer fusion case in $\boldsymbol{F_4} \sim \boldsymbol{F_1}$ receives the highest AP value of 57.2%, which we name LFCP-Basic.

Under the premise of LFCP-Basic, we further improve the SAR ship segmentation performance with the proposed SR-FRM and conduct experiments to verify its effectiveness. The results are shown in Table V. Inspired by the SR techniques [35], we concatenate the upsampled bounding box RoI feature and the pooled mask RoI feature in channel dimension and $\downarrow_{2\times}$ reduce the channel dimension of concatenated feature via 1×1 convolution, termed case1. Then, we enhance the fused feature with the proposed SSTR module, termed case2. Next, we add the residual connection from $\boldsymbol{P_2^{up}}$ for HR feature construction, termed case3. The three cases yield 0.1%, 0.2%, and 0.4% AP values compared with the baseline. Based on case3, we supplement the global context block to model the long-range dependencies inside and outside the residual connection to construct the SR-FRM. Quantitatively, LFCP-Basic receives 1% AP value increments (57.2%–58.2%) with adding the SR-FRM. Meanwhile, we remove the residual connection in SR-FRM and receive a 0.4% AP decrease, which verifies the rationality of SR-FRM architecture. In addition, we conduct experiments to select the optimal squeeze excitation ratio r in (15) and (16) of LFCP. The results of $r = 1$, $r = 2$, $r = 4$, $r = 8$, and $r = 16$ are summarized in Table VI. Quantitatively, squeeze ratio of $r = 4$ achieves the highest 58.2% AP value

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TABLE III
EFFECTS OF LFCP, HR-FIM, AND CRSB. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Model	LFCP	LFCP-Basic	HR-FIM	CRSB	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Baseline					52.2	80.6	63.7	51.8	61.3	9.9
Modules	✓				58.2	88.4	70.0	58.4	62.0	9.3
		✓	✓		58.6	88.4	71.1	58.7	63.7	14.8
		✓		✓	58.0	87.3	70.6	58.0	63.4	14.7
LFG-Net	✓		✓	✓	59.7	88.5	72.3	59.7	64.2	11.8

TABLE IV
EFFECTS OF THE NUMBER OF FPN FUSION LEVEL ON SAR SHIP INSTANCE SEGMENTATION. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Information Flow Layer	AP	AP_{50}	AP_{75}
$\boldsymbol{F_5} \sim \boldsymbol{F_2}$ $(baseline)$	52.2	80.5	63.1
$\boldsymbol{F_5} \sim \boldsymbol{F_1}$	56.8	87.2	68.8
$\boldsymbol{F_4} \sim \boldsymbol{F_1}$	**57.2**	**87.5**	**69.1**
$\boldsymbol{F_3} \sim \boldsymbol{F_1}$	55.9	85.7	67.6
$\boldsymbol{F_2} \sim \boldsymbol{F_1}$	53.5	82.2	64.6

TABLE V
ABLATION EXPERIMENTS ON SR-FRM. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Options	AP	AP_{50}	AP_{75}
baseline	57.2	87.5	69.1
case1	57.3	87.5	69.2
case2	57.4	87.3	69.6
case3	57.6	87.7	70.0
case4	57.8	87.8	**70.2**
SR-FRM	**58.2**	**88.4**	70.0

TABLE VI
SELECTION OF THE SQUEEZE RATIO. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Squeeze ratio	AP	AP_{50}	AP_{75}
r=1	57.7	88.3	69.7
r=2	57.9	88.0	**70.3**
r=4	**58.2**	**88.4**	70.0
r=8	57.8	87.9	69.9
r=16	57.6	88.0	69.8

with the 88.4% AP_{50} value and 70.0% AP_{75} value, and we set $r = 4$ in LFCP.

For the sake of verifying the efficiency of LFCP in segmenting the SAR ships, we select several state-of-the-art FPN architectures, including path aggregation FPN (PAFPN), CARAFE-FPN, recursive feature pyramid (RFP), and AugFPN [59] for comparison. As shown in Table VII, these FPNs produce an ineffectual effect on SAR ship instance segmentation as the AP value fluctuates from 51.8% to 52.2%, even lower than vanilla FPN. Meanwhile, it proves the poor transfer ability of FPNs from nature images to SAR images. In terms of the proposed LFCP, it receives an incredible 6.0% AP increments compared with vanilla FPN, which verifies the

TABLE VII
COMPARISON WITH VARIOUS FPNS. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Options	AP	AP_{50}	AP_{75}
FPN(baseline)	52.2	80.5	63.1
PAFPN	52.2	80.0	63.2
CARAFE-FPN	51.8	79.7	63.0
RFP	51.9	79.5	62.7
AugFPN	51.8	79.6	62.6
LFCP(ours)	**58.2**	**88.4**	**70.0**

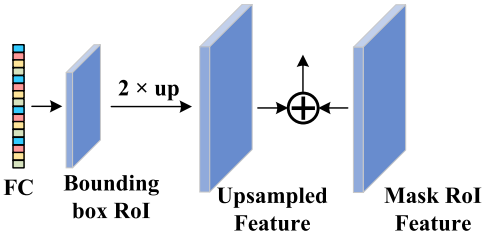

Fig. 5. Basic architecture of HR-FIM for ablation experiments.

rationality and high efficiency of the super-resolved $\boldsymbol{P_1}$ layer in capturing the regional and texture information.

2) Experiments on HR-FIM: Akin to the experiments in LFCP, we select LFCP-Basic as the baseline for elementwise validity validation of HR-FIM. Here, we select the architecture in Fig. 5 as the basic architecture for ablation experiments. The results are shown in Table VIII. Quantitatively, the AP value for the basic architecture is underwhelming and it even drops by 0.3% AP value when removing the ReLU function in the FC layer. Then, we augment the bounding box RoI feature by HR-FRM, and it receives 0.9% AP value increments (56.9% versus 57.8%), which simultaneously demonstrates the effectiveness of HR-FRM. Based on the basic architecture, we concatenate the bounding box RoI feature and mask RoI feature to build the HR RoI representations of HR-FIM, termed HR-FRM concate, and it receives 57.7% AP, 87.7% AP_{50}, and 70.1% AP_{75} value. Under the premise of HR-FRM concate, we further add an HR-FRM to capture the HR interacted RoI feature and build the HR-FIM for CRSB. Without bells and whistles, compared with the baseline, HR-FIM receives the increments of 1.4%, 0.9%, and 2% for AP, AP_{50}, and AP_{75} value, respectively. Note that the ReLU function provides 0.6% AP increments here.

3) Experiments on CRSB: As the process of the CRSB involves both compression and recovery, we conduct two groups of experiments to verify the influence of zigzag

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TABLE VIII
ABLATION EXPERIMENTS ON HR-FIM. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Options	AP	AP_{50}	AP_{75}
baseline	57.2	87.5	69.1
-ReLU	57.3	87.7	69.3
basic	56.9	87.2	68.1
basic + HR-FRM	57.8	88.2	69.8
HR-FRM concate	57.7	87.7	70.1
HR-FIM - ReLU	58.0	87.9	70.6
HR-FIM	**58.6**	**88.4**	**71.1**

TABLE IX
SELECTION OF THE MASK PREDICTION SIZE. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Zig-zag Dim	Mask Size	AP	AP_{50}	AP_{75}
900	32^2	56.3	87.1	67.9
	64^2	57.3	87.3	69.1
	128^2	**58.0**	**87.1**	**70.3**
	256^2	57.7	86.7	70.2
	512^2	57.3	85.6	69.9

TABLE X
SELECTION OF THE COMPRESSION DIMENSION. ALL THE RESULTS ARE EVALUATED ON THE HRSID TEST SET

Mask Size	Dim	AP	AP_{50}	AP_{75}
128	500	57.8	87.2	69.7
	700	**58.0**	**87.3**	**70.6**
	900	58.0	87.1	70.3
	1100	57.9	87.1	70.3
	1300	58.0	87.1	70.5
	1500	57.6	87.2	70.1

compression dimension and the recovered mask size on the SAR ship segmentation performance, respectively. As shown in Table IX, we select the 900 dimensions of the output FC layer as the baseline to verify the effect of the recovered size, including 32^2, 64^2, 128^2, and 256^2 pixels, of the image. Quantitatively, the recovered image with the size of 128^2 pixels achieves the highest 58.0% AP value (1.7% AP increments to the 32^2 pixels case) with regard to all cases. For the sake of selecting the optimal compression dimension for zigzag compression, we conduct experimental verification. The results are shown in Table X, where the compressed dimension of 500, 700, 900, 1100, 1300, and 1500 is used for comparison. It is obvious that the segmentation performance of the zigzag dimension still fluctuates from 57.6% AP value to 58.0% AP value. Therefore, we select the 700 zigzag dimension that achieves the top performance for our CRSB.

E. Comparative Results

Elevated from verifying the intranetwork to internetwork effectiveness of the proposed LFG-Net, we select state-of-the-art instance segmentation network, including the top-down methods (including Yolact and SOLO), bottom-up methods (including Mask R-CNN, Pointrend, Instaboost [60], and R-ARE-Net), and the derived cascaded methods (Cascade Mask R-CNN, HTC, and SCNet) for comparison. In the absence of special instruction, all the methods adopt the backbone of ResNet-50 and the default hyperparameter settings in [65]. With the MS COCO evaluation metrics defined in Section IV-B, we comprehensively compare the quantitative results of LFG-Net with state-of-the-art methods in AP, threshold AP (including AP_{50} and AP_{75}), and scale AP (including AP_S, AP_M, and AP_L). Meanwhile, the network complexity and speed of the trained model are calculated by parameters and FPS, respectively. As for the proposed LFG-Net, we supplement advanced backbone networks or techniques, e.g., ResNeXt and multiscale training, for further improving its performance. Quantitative results are shown in Table XI. Without bells and whistles, LFG-Net achieves 59.7% AP, 88.5% AP_{50}, and 72.3% AP_{75} value. Compared with HTC, LFG-Net receives 5.3% AP increments under the condition that the number of parameters merely increased by 22.49 M and FPS is reduced by 2. In addition, we enhance LFG-Net with the advanced ResNeXt-64 $\times$ 4d backbone network and the multiscale training technique with the image size of 800×800 and 800×1000 pixels. As expected, LFG-Net with ResNeXt-64 $\times$ 4d and multiscale training receives 0.5% and 3.2% AP increments, respectively. When ResNeXt-64 $\times$ 4d and multiscale training technique are simultaneously adopted in training LFG-Net, termed LFG-Net*, it receives 63.9% AP, 90.1% AP_{50}, and 76.8% AP_{75} value, which is the highest quantitative result among the state-of-the-art methods. Compared with Mask R-CNN baseline, LFG-Net* receives incredible 11.7%, 9.5%, and 13.1% increments in AP, AP_{50}, and AP_{75}, respectively. As for the scale AP, LFG-Net* receives 11.8% and 8.2% increments in AP_S and AP_M, respectively, where the incremental trend of AP_S, which represents the small SAR ships occupying the majority of the objects, is similar to that of AP.

In the internetwork comparison, we group them into three parts for analysis. Note that due to the small object characteristics of ships in HR SAR images, large ships occupy a low proportion (2% ships) in HRSID, and insufficient training samples reduced the AP_L of state-of-the-art methods. First, in the absence of RoI, where the location of the ship is preconceived, in top-down instance segmentation methods, bottom-up methods encounter difficulties in directly segmenting the ships under the complex background signal noise in SAR images. SOLO and Yolact receive the unsatisfactory 13.8% and 35.3% AP values, respectively. Second, based on the Mask R-CNN baseline, Instaboost, DCN [62], CARAFE, GRoIE [61], Mask Scoring R-CNN, and PointRend provide the data augmentation technique, advanced backbone network, advanced FPN fusion and mask prediction upsampling, novel RoI extractor, mask scoring head, and the instance segmentation rendering paradigm, respectively. However, most of the advanced methods suffer difficulties in segmenting the SAR ships, and the AP value fluctuates from 51.6% to 53.8% with competitive parameters and FPS. Among them, PointRend obtains a promising 53.8% AP and 52.6% AP_S value with point-based segmentation predictions. Since R-ARE-Net is dedicated to instance segmentation in the complex background of aerial images, instead of considering the small object

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TABLE XI
INSTANCE SEGMENTATION RESULTS ON THE HRSID TEST SET

Type	Model	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params.	FPS
Bottom-up Methods	*SOLO* [28]	13.8	27.8	13.8	14.2	13.6	3.8	54.92M	14.2
	Yolact [27]	35.3	67.2	35.2	34.1	50.0	6.2	34.73M	17.3
Top-Down Methods	*Mask R-CNN* [21]	52.2	80.6	63.7	51.8	61.3	9.9	43.75M	14.9
	Instaboost [60]	51.6	79.9	62.6	51.3	59.5	10.8	43.75M	14.6
	PointRend [26]	53.8	81.8	65.3	53.1	63.3	15.5	55.53M	12.3
	GRoIE [61]	52.0	79.7	62.8	51.4	61.1	17.5	47.54M	8.3
	CARAFE [32]	51.7	79.6	62.8	51.4	59.2	12.7	49.35M	13.5
	DCN [62]	51.6	79.2	62.7	51.0	61.5	13.8	47.23M	13.8
	Mask Scoring R-CNN [24]	53.0	80.8	63.8	52.6	61.0	11.8	60.01M	14.7
	R-ARE-Net [18]	53.6	80.4	65.9	55.3	55.2	13.5	46.58M	10.4
Cascaded Methods	*QueryInst* [63]	44.2	69.5	53.2	43.4	54.6	12.2	172.22M	4.5
	Cascade Mask R-CNN [22]	53.3	82.0	64.0	52.7	61.9	18.3	76.08M	13.5
	Hybrid Task Cascade [23]	53.6	82.3	64.7	52.8	63.2	18.6	79.73M	9.7
	Detectors [64]	54.1	82.4	65.5	53.3	64.2	20.7	134.00M	6.3
	SCNet [41]	54.4	82.4	65.9	54.1	62.1	13.2	94.29M	8.6
Ours	*LFG-Net*	59.7	88.5	72.3	59.7	64.2	11.8	116.78M	6.6
	LFG-Net+ResNeXt	60.2	89.3	72.8	59.9	66.2	19.1	174.28M	5.0
	LFG-Net+mstrain	62.9	89.8	75.7	62.6	68.9	24.1	116.78M	6.6
	*LFG-Net**	**63.9**	**90.1**	**76.8**	**63.6**	**69.5**	**42.5**	174.28M	5.0

TABLE XII
INSTANCE SEGMENTATION RESULTS ON THE PSEG-SSDD TEST SET (35% IMAGES)

Type	Model	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params.	FPS
Bottom-up Methods	*SOLO*	14.7	31.0	14.1	9.9	31.3	9.1	54.92M	27.7
	Yolact	41.7	82.8	35.2	40.2	50.1	30.0	34.73M	32.9
Top-Down Methods	*Mask R-CNN*	57.9	90.2	72.3	58.0	59.2	16.6	43.75M	24.0
	Instaboost	57.3	88.5	71.3	57.4	58.6	8.4	43.75M	25.6
	PointRend	57.7	90.1	72.6	57.4	60.2	18.3	55.53M	19.9
	GRoIE	57.2	89.2	72.7	57.3	59.2	21.4	47.54M	10.6
	CARAFE	54.1	85.7	66.4	55.1	54.3	13.3	49.35M	21.8
	DCN	56.7	89.2	69.9	56.5	59.4	16.8	47.23M	23.2
	Mask Scoring R-CNN	57.5	88.4	72.1	56.8	60.6	21.4	60.01M	25.6
	R-ARE-Net	58.7	89.5	74.0	59.0	60.7	8.2	46.58M	8.1
Cascaded Methods	*QueryInst*	48.1	75.4	59.2	46.9	52.8	24.9	172.22M	9.1
	Cascade Mask R-CNN	57.9	89.3	72.5	57.5	60.8	16.7	76.80M	18.5
	Hybrid Task Cascade	58.3	90.0	73.2	57.7	61.3	21.9	79.73M	14.8
	Detectors	59.3	90.8	73.9	58.7	61.8	17.1	134.00M	7.9
	SCNet	56.9	87.8	71.6	57.4	58.2	23.8	94.29M	10.9
Ours	*LFG-Net**	**64.2**	**95.0**	**81.1**	**63.1**	**68.2**	**43.1**	176.34M	10.1

characteristics of SAR ships, it receives 53.6% AP value. Third, the cascaded methods are derived from the top-down methods but adopt the cascaded mask prediction architecture to improve their performance, which simultaneously brings the increase of parameters and the drop of FPS. Quantitatively, most cascaded methods achieve 53.3% AP value. However, even SCNet with the highest 54.4% AP merely receives 2.2% AP increments compared with Mask R-CNN. Nevertheless, compared with SCNet, the proposed LFG-Net* still receives an incredible 9.5% AP increments, which verifies the effectiveness of LFG-Net*.

In addition to the quantitative results, we visualize the qualitative SAR ship instance segmentation results in Fig. 6. With regard to the adjacent distributed ships in column 1, state-of-the-art methods tend to produce missing segmentations (denoted as yellow rectangles), aliasing masks (denoted as red rectangles), false alarms (denoted as blue rectangles), and undersegmentations (denoted as green rectangles) due to the strong interference from adjacent ship feature in the RoI. As for the large aspect ratio of ships in column 2 and the strong reflections provided by artificial objects in the port in column 3, state-of-the-art methods still produce unsatisfactory results. In terms of the small ships in columns 4 and 5, state-of-the-art methods are prone to generating missing segmentations as the higher level feature maps ignore the regional and texture information of ships even the ships in the SAR images preserve the details. In the counterpart results of LFG-Net*, the background interference and feature interference in adjacent distributed ships are suppressed by the HR techniques in the modules, and the contour detail of ships fits the ground truth perfectly. Besides, missing segmentations of small ships in the canal are handled by the extended LFCP $\boldsymbol{P_1}$ level.

For the sake of verifying the generalization ability of LFG-Net*, we conduct comparative experiments in

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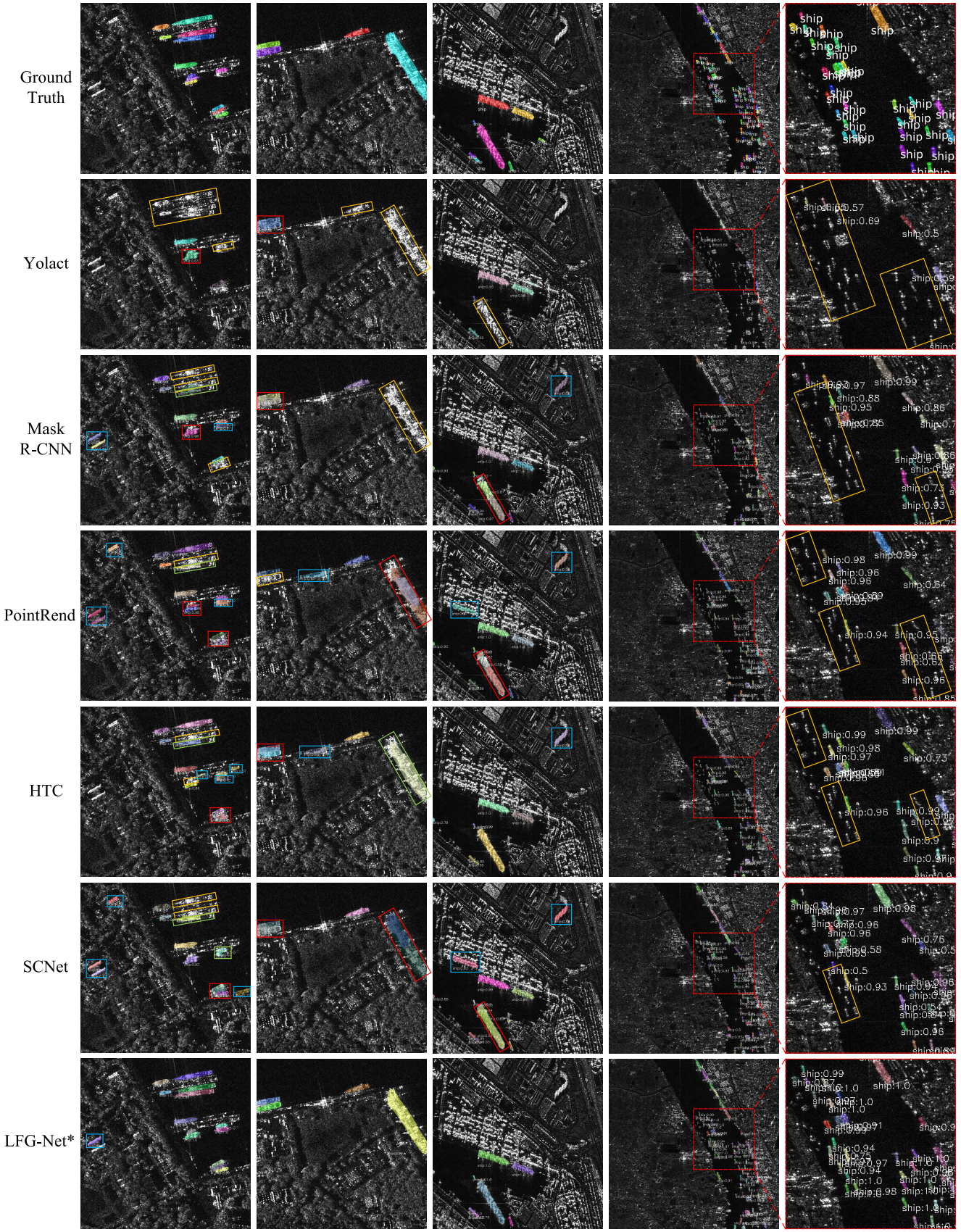

Fig. 6. Qualitative results of LFG-Net* and state-of-the-art methods on HRSID. Row 1 is the ground truth; rows 2–7 are the visualized instance segmentation results of Yolact, Mask R-CNN, PointRend, HTC, SCNet, and LFG-Net*, respectively. Red rectangles are the aliasing masks. Blue rectangles are the false alarms. Yellow rectangles are the missing segmentations. Green rectangles are the undersegmentations. The red dotted square represents locally magnified details of the small ship instances.

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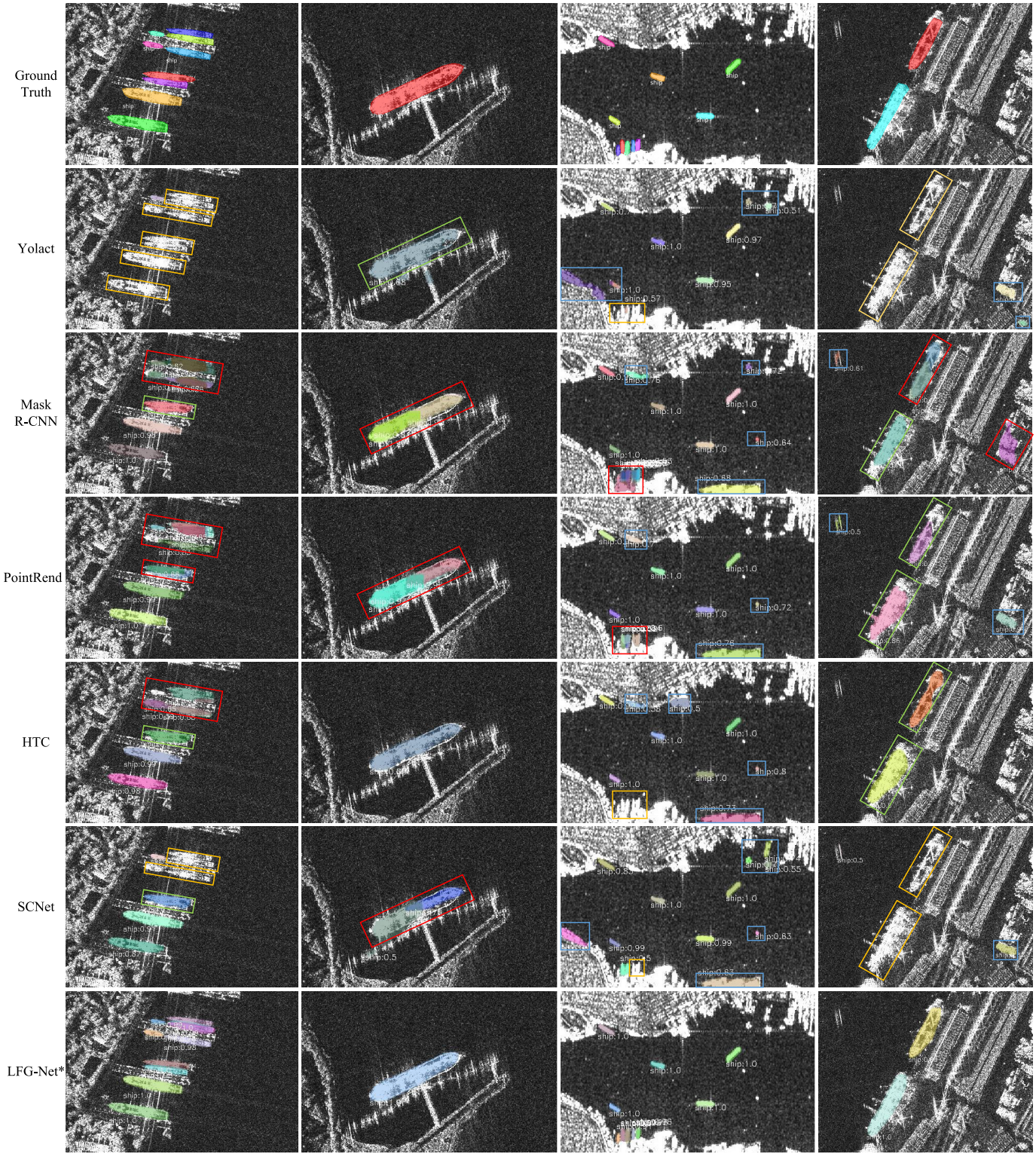

Fig. 7. Qualitative results of LFG-Net* and state-of-the-art methods on PSeg-SSDD. Row 1 is the ground truth; rows 2–7 are the visualized instance segmentation results of Yolact, Mask R-CNN, PointRend, HTC, SCNet, and LFG-Net*, respectively. Red rectangles are the aliasing masks. Blue rectangles are the false alarms. Yellow rectangles are the missing segmentations. Green rectangles are the undersegmentations.

PSeg-SSDD. Similarly, we employ the same state-of-the-art methods as that of HRSID for comparison. Note that the image size of multiscale training is adjusted to 512×512 and 512×640 pixels. Quantitative results are summarized in Table XII. As expected, our LFG-Net* achieves the highest 64.2% AP, 95.0% AP_{50}, and 81.1% AP_{75} value. As for the scale AP, it achieves the 63.1% AP_S, 68.2% AP_M, and 43.1% AP_L value. Distinguished from the results in HRSID, the top-down instance segmentation results in PSeg-SSDD fluctuate from 54.1% to 57.9% AP value, with the Mask R-CNN

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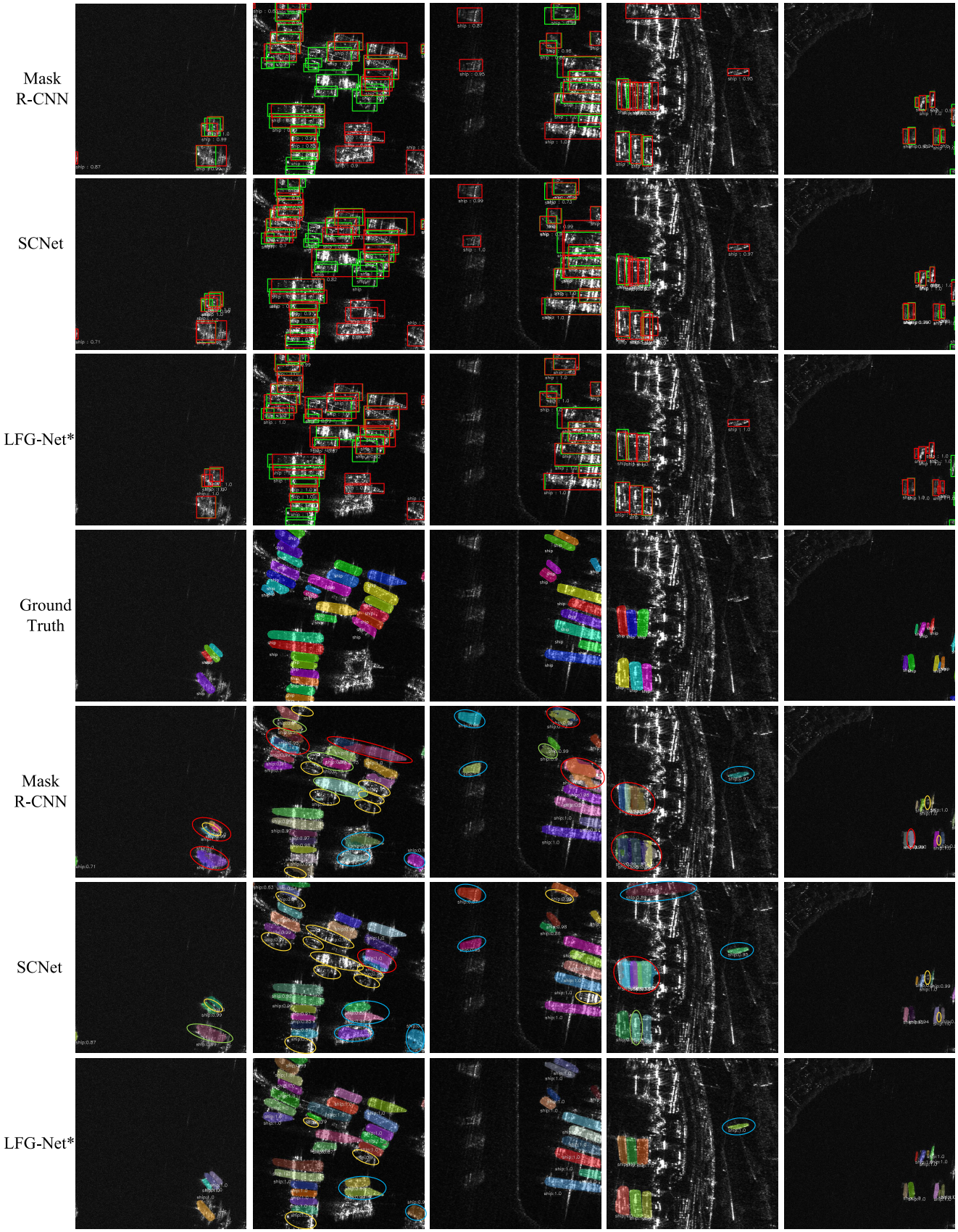

Fig. 8. Qualitative ship detection and ship instance segmentation results of Mask R-CNN, SCNet, and LFG-Net* on the AirSARShip dataset. Rows 1–3 are the ship detection results, where red bounding boxes and green bounding boxes are the predicted results and ground truth, respectively. Row 4 is the ground-truth mask. Rows 5–7 are the instance segmentation results, where red ellipses are the aliasing masks, blue ellipses are the false alarms, yellow ellipses are the missing segmentations, and green rectangles are the undersegmentations.

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TABLE XIII
SHIP DETECTION AND SHIP INSTANCE SEGMENTATION RESULTS ON THE AIR-SARSHIP TEST SET

Model	*Ship Detection*						*Ship Instance Segmentation*						Gap	Params.	FPS
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L			
Mask R-CNN	56.8	82.2	64.0	49.4	61.9	25.7	49.1	77.1	56.9	40.1	53.3	30.8	7.7	43.75M	21.8
Instaboost	56.0	83.2	63.1	45.8	61.7	30.2	48.1	78.2	54.8	37.0	52.6	32.2	7.9	43.75M	23.0
PointRend	58.3	83.4	67.1	50.0	63.7	29.7	54.1	80.5	64.0	41.6	59.0	40.9	4.2	55.53M	20.1
GRoIE	57.5	82.0	66.3	49.2	62.9	28.2	51.4	78.7	59.9	40.7	55.8	37.2	6.1	47.54M	10.3
CARAFE	56.9	82.7	64.6	49.7	61.9	30.6	50.1	78.6	58.0	39.4	54.4	35.9	6.8	49.35M	19.0
DCN	56.3	82.6	63.9	47.6	61.8	29.8	48.6	77.4	55.3	38.2	52.8	34.1	7.7	47.23M	19.6
Mask Scoring R-CNN	58.0	83.1	66.1	50.0	63.0	32.2	49.4	77.6	56.5	39.3	53.6	34.0	8.6	60.01M	20.8
R-ARE-Net	56.6	83.3	64.8	49.0	61.9	31.5	53.8	80.4	63.8	46.4	58.5	32.2	2.8	46.58M	12.1
QueryInst	40.1	64.5	42.8	37.3	43.1	25.4	35.4	60.3	38.3	29.1	38.4	31.7	4.7	172.22M	6.4
Cascade Mask R-CNN	60.6	83.3	69.2	50.8	66.0	34.1	50.9	78.4	58.5	40.0	55.4	34.3	9.7	76.80M	18.0
Hybrid Task Cascade	60.7	84.1	69.0	50.9	66.1	36.3	52.7	80.2	61.3	41.5	57.0	39.8	8.0	79.73M	13.6
Detectors	61.7	85.0	69.2	51.5	67.1	37.5	54.5	81.5	63.6	42.6	58.9	42.6	7.2	134.00M	7.7
SCNet	60.1	83.2	67.7	50.8	65.6	32.9	54.3	80.6	63.5	42.8	58.7	42.5	5.8	94.29M	10.5
*LFG-Net**	**64.8**	**84.1**	**73.6**	**58.8**	**69.3**	**39.2**	**61.8**	**82.1**	**70.8**	**53.8**	**65.3**	**53.4**	3.0	174.28M	9.0

baseline reaching the highest AP value of 57.9%. In particular, R-ARE-Net achieves the 0.8% AP increments and receives the 58.7% AP value when migrating to SAR images. As for the cascaded methods, the detectors receive the highest 59.3% AP value. Still, insufficient training samples (2.7%) in PSeg-SSDD reduced the AP_L of state-of-the-art methods.

Qualitatively, we visualized the SAR ship instance segmentation results in Fig. 7. Similarly, state-of-the-art methods still encounter the cases of aliasing masks, false alarms, missing segmentations, and undersegmentations. The relatively low resolution in PSeg-SSDD leads to severe false alarms and aliasing masks, as shown in column 3 of Fig. 7. Moreover, ships in the port tend to produce undersegmentations and missing segmentations. In the counterpart results of LFG-Net*, these cases are suppressed to a certain extent, and the segmented masks of adjacent distributed ships, port ships, and so on fit the contours of ships well.

In addition, we conduct experiments on the Air-SARShip dataset to simultaneously verify the ship detection and ship instance segmentation performance of LFG-Net* compared with the top-down and cascaded methods (simultaneously supports detection and segmentation tasks). Note that the image size of multiscale training is adjusted to 600×600 and 600×800 pixels. Qualitative results are summarized in Table XIII. As for ship detection, top-down methods fluctuate from 56.0% to 58.3% detection AP value. Notably, the cascade detection architecture in cascaded methods facilitates the generation of accurate detection boxes and thus improves the ship detection AP, where detectors achieve 4.9% detection AP increments compared with Mask R-CNN. Among them, our LFG-Net* achieves the highest 64.8% detection AP value. As for ship instance segmentation, both top-down and cascaded methods encounter difficulties in accurately segmenting the ships, where the AP fluctuates from 35.4% AP value to 54.5% AP value. Notably, our LFG-Net* achieves the highest 61.8% AP, 82.1% AP_{50}, and 70.8% AP_{75} value. Consistent with the trend in HRSID, the highest AP_S value of general methods in CV is 42.8% in SCNet. As R-ARE-Net is transferred from optical remote sensing images to SAR images, which considers the small object characteristics of small objects, it receives 46.4% AP_S value and 58.5% AP_M value. Notably, the proposed LFG-Net* guided by low-level feature receives the highest 53.8% AP_S, 65.3% AP_M, and 53.4% AP_L value. Considering that accurate ship detection can improve the segmentation performance, we supplemented the gap of detection AP and segmentation AP of the methods in Table XIII. Obviously, the AP gap of 3.0% in LFG-Net* is 4.7% lower than that of Mask R-CNN.

Qualitatively, we visualized the ship detection and ship instance segmentation results of Mask R-CNN, SCNet, and LFG-Net* in Air-SARShip in Fig. 8. Rows 1–3 are the ship detection results, row 4 is the ground-truth mask, and rows 5–7 are the ship instance segmentation results. With regard to the small ships in columns 1 and 5, Mask R-CNN tends to produce missing segmentations (denoted as yellow ellipses) and detections, aliasing masks (denoted as red ellipses); SCNet tends to produce undersegmentations (denoted as green ellipses). As for the adjacent distributed ships in columns 2 and 3, Mask R-CNN and SCNet exist severe false alarms (denoted as blue ellipses) and missing segmentations. In the complex background of column 4, Mask R-CNN and SCNet tend to produce aliasing masks and false alarms. Notably, our LFCP* can suppress these defects and the segmentation results fit the contour well.

V. CONCLUSION

Interpreting ships in SAR images is a challenging task due to the grayscale image characteristic and signal noise interference in SAR images. In previous work, researchers tended to utilize horizontal or oriented bounding boxes to detect ships. However, the refined contour features and the size of ships are ignored at the same time. In this article, we introduce the notion of instance segmentation in the SAR field and propose the dedicated network, termed LFG-Net.

Compared with the objects in natural images, the majority of ships retain the small size in the SAR images. Therefore, we tend to extend the lower feature level of FPN to detect and segment the small ships and supplement the SR techniques to

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enhance the low-level $\boldsymbol{P_1}$ layer, termed LFCP. Moreover, the HR-FIM and CRSB interact with the bounding box RoI and mask RoI and produce the super-resolved mask prediction, respectively. With the dedicated LFCP, HR-FIM, and CRSB, LFG-Net achieves the 59.7% AP, 88.5% AP_{50}, and 59.7% AP_S value. Without bells and whistles, compared with the Mask R-CNN baseline, the improved LFG-Net* achieves 11.7%, 6.3%, and 12.7% AP increments on the HRSID test set, PSeg-SSDD test set (35% images), and Air-SARShip test set (35% images), respectively. Moreover, the visualized instance segmentation results indicate that LFG-Net* is capable of segmenting the adjacent distributed ships, ships with large aspect ratios, ships with strong reflection noise interference, and small ships. Besides, the missing segmentations, aliasing masks, false alarms, and undersegmentations can be suppressed to a certain extent.

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